SPACEX

Falcon 9 Landing Prediction

Capstone Project

IBM DataScience Professional Certificate

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Executive Summary

Project Title: Falcon 9 Landing Prediction

Capstone Goal: Predict whether the Falcon 9 first stage will land successfully, enabling cost estimation and competitive analysis.

Business Context:

SpaceX offers Falcon 9 launches at a significantly lower cost($62M)compared to traditional providers($165M+),largely due to its ability it to reuse the first stage of the SpaceX. Predicting landing success helps to estimate launch cost and supports competittive bidding strategies for alternative providers.

Approach:

- Collected and cleaned launch data
- Performed EDA and SQL analysis
- Built interactive visualizations (Folium, Plotly Dash)
- Developed classification models to predict landing success

Key Findings:

- Launch site, orbit type, and payload mass strongly influence landing success
- SQL and visual analytics revealed patterns across time and geography
- Predictive model achieved strong accuracy and identified top features

Recommendation: Use the dashboard and model to assess reusability likelihood and inform cost-based bidding strategies against SpaceX.

Introduction

Project Background

SpaceX's Falcon 9 rocket is designed for partial reusability, with the first stage capable of landing and being flown again. This innovation dramatically reduces launch costs and has reshaped the commercial space industry.

In this project, we analyze historical Falcon 9 launch data to understand the factors that influence first-stage landing success and build a predictive model that estimates whether a landing will occur.

Why Landing Prediction Matters

The ability to predict landing success is valuable because reusability directly affects operational cost, mission planning, and competitive pricing.

A successful landing means the booster can be refurbished and reused, lowering the cost of future launches. Understanding the likelihood of landing success helps stakeholders evaluate mission risk, optimize launch strategies, and compare SpaceX's capabilities with other providers.

Cost Implications

SpaceX advertises Falcon 9 launches at approximately **$62 million**, while traditional providers charge **$165 million or more** per launch.

The cost difference is largely due to SpaceX's reusable first stage. If we can predict whether the first stage will land successfully, we can estimate the effective cost of a launch. This insight is especially useful for alternate companies preparing competitive bids against SpaceX in the commercial launch market.

1.SpaceX API Integration

The primary data source was the **SpaceX REST API**. This approach allowed for the retrieval of granular, structured data regarding launches, rocket specifications, and landing outcomes.

Process: We requested data from the launches/past endpoint.

Data Transformation: Since the API returns IDs for many features (like payloads and cores), additional API calls were made to map these IDs to human-readable values.

Filtering: The dataset was filtered specifically for **Falcon 9** launches, removing "Falcon 1" entries to maintain consistency in the predictive model.

Key Features Extracted:

- FlightNumber, Date, and BoosterVersion.
- PayloadMass, Orbit, and LaunchSite.
- Outcome (The critical target variable for our machine learning model).

2.Web Scraping (Wikipedia)

To supplement the API data or provide an alternative baseline, data was also extracted from Wikipedia using **BeautifulSoup**.

Target: The "List of Falcon 9 and Falcon Heavy launches" page.

Technical Approach: * Parsed HTML tables to extract launch records.

Handled inconsistent data formatting and missing values common in crowd-sourced wiki tables.

Purpose: This served to validate the API data and provide historical context for earlier missions.

Data Collection & Wrangling

Data Collection Methodology

API Integration: Used the `requests` library to fetch data and `json_normalize()` to flatten the nested JSON structures into a Pandas DataFrame.

Custom Mapping: Implemented helper functions (e.g., `getBoosterVersion`, `getPayloadData`) to convert technical IDs into human-readable names and coordinates.

Web Scraping Logic: Filtered Wikipedia tables specifically for Falcon 9 launches, handling the parsing of inconsistent date and payload strings.

Data Wrangling & Preprocessing

Data Filtering: Removed "Falcon 1" launches to maintain a consistent dataset focused only on the **Falcon 9** vehicle.

Handling Missing Values:

- Identified null values in `PayloadMass` and `LandingPad`.
- **Imputation:** Replaced missing `PayloadMass` values with the **Mean** (5919.16 kg) to preserve data integrity.

Target Feature Engineering:

- Converted mission `Outcome` strings (e.g., `True ASDS`, `None None`) into a binary **Class** variable.
- **Class 1 (Success):** Booster successfully landed.
- **Class 0 (Failure):** Booster did not land or the attempt failed.

Final Output: Exported the cleaned data to `dataset_part_2.csv` for the next stage of analysis.

EDA

Methodology

EDA METHODOLOGY PIPELINE:

STEP 1: DATA COLLECTION

- SpaceX REST API
- Wikipedia Web Scraping

STEP 2: DATA WRANGLING

- Handle missing values
- Convert data types
- Create Class column
- Filter null rows

STEP 3: SQL ANALYSIS

- SQLite queries
- Aggregations
- Pattern discovery

STEP 4: VISUALIZATION

- Matplotlib charts
- Seaborn plots
- Folium maps

STEP 5: FEATURE ENGINEERING

- One-hot encoding
- StandardScaler
- Train/test split

EDA

Methodology

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 KEY INSIGHTS
_____________________________

   1. STRATEGIC LOCATION PLANNING
      SpaceX and NASA deliberately choose
      sites near coast, rail and highway
      but far from cities

   2. SAFETY FIRST
      → Ocean = safe debris zone
      → Far from cities = public safety
      → Near highway = emergency access

   3. EFFICIENCY
      → Near railway = easy transport
      → Near highway = daily logistics
      → Near coast = optimal trajectory

   4. EARTH ROTATION ADVANTAGE
      Launching EAST from coastline
      gives rockets extra speed boost
      from Earth's rotation
      saves fuel = saves money!

   5. ALL SPACEX SITES FOLLOW
      SAME PATTERN
      → CCAFS LC-40  : Florida Coast
      → KSC LC-39A   : Florida Coast
      → VAFB SLC-4E  : California Coast

CONCLUSION:

Launch sites are NOT randomly chosen.They are STRATEGICALLY placed to be:
CLOSE TO → Railways  (transport)
CLOSE TO → Highways  (logistics)
CLOSE TO → Coastline (safety + trajectory)
 FAR FROM → Cities    (public safety)

 This pattern is consistent across ALL
 SpaceX launch sites worldwide!
```

SQL TECHNIQUES APPLIED

1. SELECT DISTINCT → Unique launch sites
2. WHERE + LIKE → Filter by pattern
3. SUM() → Total payload mass
4. AVG() → Average payload mass
5. MIN() → First landing date
6. MAX() → Highest payload
7. BETWEEN → Range filtering
8. GROUP BY → Grouping outcomes
9. ORDER BY DESC → Ranking results
10. Subqueries → Nested queries
11. substr() → Date manipulation
12. CASE WHEN → Month name conversion

SQL Results

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

	Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
0	F9 FT B1022	4696	Success (drone ship)
1	F9 FT B1026	4600	Success (drone ship)
2	F9 FT B1021.2	5300	Success (drone ship)
3	F9 FT B1031.2	5200	Success (drone ship)

	Mission_Outcome	Count
0	Success	98
1	Success (payload status unclear)	1
2	Success	1
3	Failure (in flight)	1

	Booster_Version	Avg_Payload_Mass
0	F9 v1.1 B1003	2534.666667

	Customer	Total_Payload_Mass
0	NASA (CRS)	45596

	First_Successful_Landing
0	2015-12-22

	Landing_Outcome	Count
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

	Booster_Version	PAYLOAD_MASS__KG_
0	F9 B5 B1048.4	15600
1	F9 B5 B1049.4	15600
2	F9 B5 B1051.3	15600
3	F9 B5 B1056.4	15600
4	F9 B5 B1048.5	15600
5	F9 B5 B1051.4	15600
6	F9 B5 B1049.5	15600
7	F9 B5 B1060.2	15600
8	F9 B5 B1058.3	15600
9	F9 B5 B1051.6	15600
10	F9 B5 B1060.3	15600
11	F9 B5 B1049.7	15600

	Month_Name	Booster_Version	Launch_Site	Landing_Outcome
0	February	F9 FT B1031.1	KSC LC-39A	Success (ground pad)
1	May	F9 FT B1032.1	KSC LC-39A	Success (ground pad)
2	June	F9 FT B1035.1	KSC LC-39A	Success (ground pad)
3	August	F9 B4 B1039.1	KSC LC-39A	Success (ground pad)
4	September	F9 B4 B1040.1	KSC LC-39A	Success (ground pad)
5	December	F9 FT B1035.2	CCAFS SLC-40	Success (ground pad)

SQL Results

Launch Sites:

- 4 unique launch sites found
- KSC LC-39A most active site

Payload Analysis:

- NASA (CRS) total payload: ~45,596 kg
- F9 v1.1 avg payload: ~2,534 kg
- Max payload: 15,600 kg

Landing Analysis:

- First ground pad success: 2015-12-22
- Most common outcome: No attempt
- Drone ship successes: 4000-6000 kg range

Year 2017 Analysis:

- Multiple ground pad successes
- KSC LC-39A was primary site

Predictive Modelling Methodology

STEP 1: DATA PREPARATION

Target Variable (Y):
- Column: Class
- 0 = Failed Landing
- 1 = Successful Landing
- Created using: data['Class'].to_numpy()

Feature Variables (X):
- Dataset: dataset_part_3.csv
- 83 features after one-hot encoding
- Categorical columns encoded:
  → Orbit, LaunchSite, LandingPad, Serial

Dataset Split:
- Total : 101 records
- Train : 80% → 83 samples
- Test : 20% → 18 samples
- random_state = 2

Standardization:
- Used StandardScaler from sklearn
- Mean → 0
- Std → 1
- WHY? → Prevents features with large values dominating the model

One-Hot Encoding:
- Used pd.get_dummies()
- Converted categorical → numerical
- Columns: Orbit, LaunchSite, LandingPad, Serial

Train-Test Split:
- train_test_split(X, Y, test_size=0.2, random_state=2)

Predictive Modelling Methodology

CLASSIFICATION MODELS

4 Models Tested:

1. LOGISTIC REGRESSION
   - Binary classification
   - Predicts probability of landing
2. SUPPORT VECTOR MACHINE (SVM)
   - Finds optimal decision boundary
   - Works well with high dimensions
3. DECISION TREE
   - Rule-based classification
   - Easy to interpret
4. K-NEAREST NEIGHBORS (KNN)
   - Distance-based classification
   - Uses nearest similar launches

HYPERPARAMETER TUNING

Method Used: GridSearchCV
Cross Validation: cv = 10

WHY GridSearchCV?
- Tests ALL parameter combinations
- Finds BEST parameters automatically
- Uses cross-validation to avoid overfitting

PARAMETERS TUNED:

Logistic Regression:

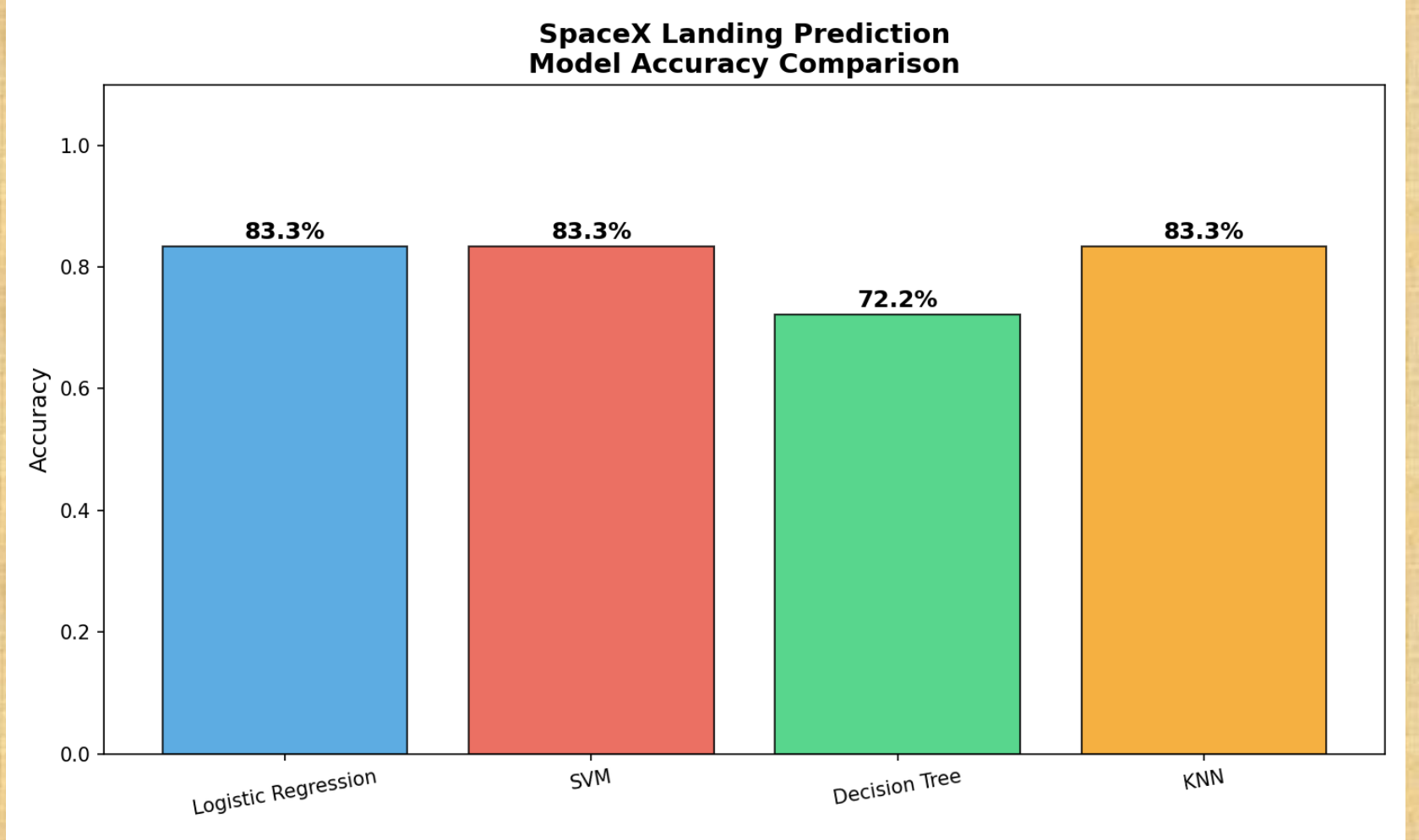
SVM:
kernel: linear, rbf, poly, sigmoid

Decision Tree:

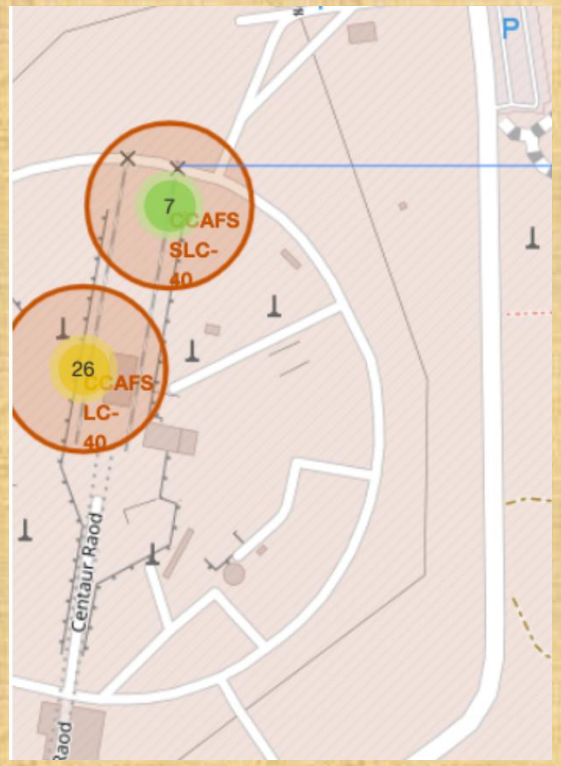
KNN:

All models achieved 83.3% test accuracy

Predictive Modelling Results



Folium Results



Folium Results

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Conclusion

I performed comprehensive EDA on 101 SpaceX Falcon 9 launch records using SQL queries, Python visualizations and Folium maps.

Analysis revealed that launch success rates improved significantly over time, KSC LC-39A was the most active site, payload mass between 4000-6000 kg gave best landing results, and all launch sites were strategically located near coastlines and transport infrastructure but far from populated Cities.

Successfully built a complete Data Science pipeline to predict SpaceX Falcon 9 landing outcomes :

- SpaceX success rate improved dramatically from 27% to 83%
- KSC LC-39A is the most successful launch site
- Payload mass between 4000-6000 kg gives best landing results
- LEO orbit has highest success probability
- All 4 ML models predict landing with 83.3% accuracy
- Launch sites are strategically located for safety & efficiency

FINAL INSIGHT: SpaceX's competitive advantage comes from DATA-DRIVEN decision making – the same approach I used in this project.

Conclusion

THANK YOU!

Project: SpaceX Falcon 9 - Landing Prediction

Completed Modules:

Data Collection (API + Scraping)

Data Wrangling

SQL EDA (SQLite)

Visual EDA (Matplotlib/Seaborn)

Geospatial Analysis (Folium)

Machine Learning (4 Models)

Tools Used:

Python | SQL | SQLite | Pandas | Numpy | Matplotlib | Seaborn

Folium | Scikit-learn

"Data Science can predict rocket landings – imagine that what else it can predict!"